

Compensation application and wild-boar tolerance

Statistical consulting for Mengxi Kou

17 September 2026

Compensation application and tolerance

- ▶ Wild-boar conflict in Giant Panda National Park, China
- ▶ Fresh analysis for Mengxi Kou
- ▶ Measurement, associations, mechanisms, and robustness

The substantive question

- ▶ Can financial compensation support coexistence?
- ▶ Do applicants report different tolerance?
- ▶ Which lived experiences and institutional contacts accompany that difference?

Study design and retained sample

- ▶ 625 households: Baoxing 160; Qingchuan 465
- ▶ Different interview periods: August 2024 vs March–April 2025
- ▶ 25 townships; 177 complete village keys; 49 earlier exclusions unexplained

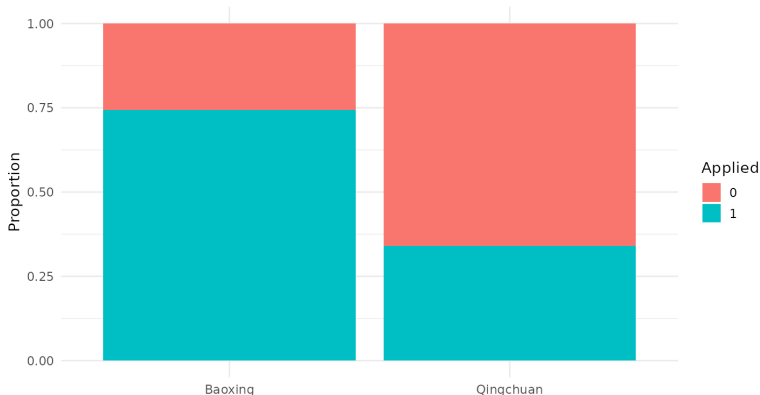
Application is not verified payment

- ▶ CEO asks whether the respondent applied
- ▶ 277 applicants; 348 non-applicants
- ▶ Receipt, timing, and eligibility are not independently established

Application differs by county

- ▶ County adjustment is essential context, not causal identification.

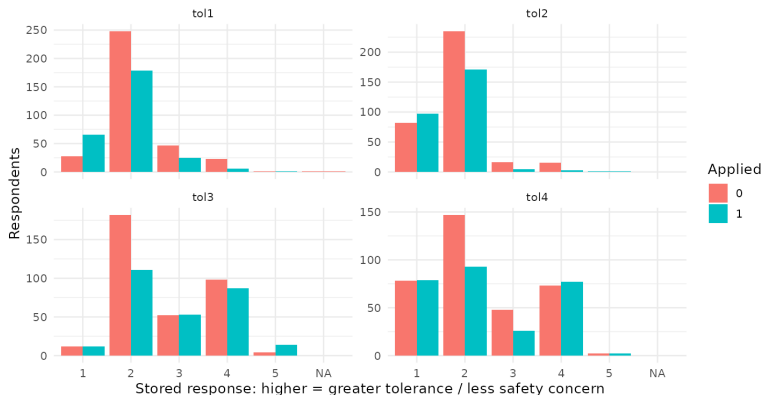
Application rates differ strongly by county



Four questions, four dimensions

- Non-lethal management; population control; safety; accepting damage

Four distinct tolerance questions

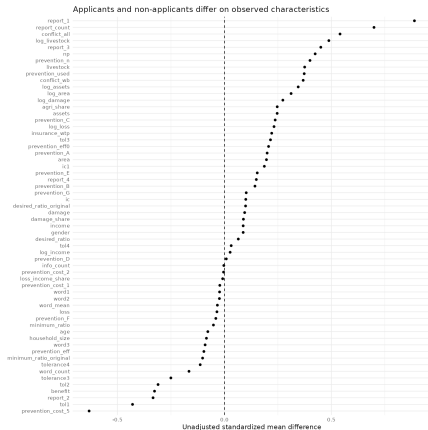


A traceable data rebuild

- ▶ Preserve the workbook; record every derived change
- ▶ Restore raw nonresponse and missing-word zeros
- ▶ Rebuild institutional counts; flag unresolved percentages and units

EDA establishes the comparison

- ▶ Profiles, missingness, skewness, ceiling effects, and imbalance
- ▶ Raw questionnaire fields retain additional substantive information



An analysis map, not a model contest

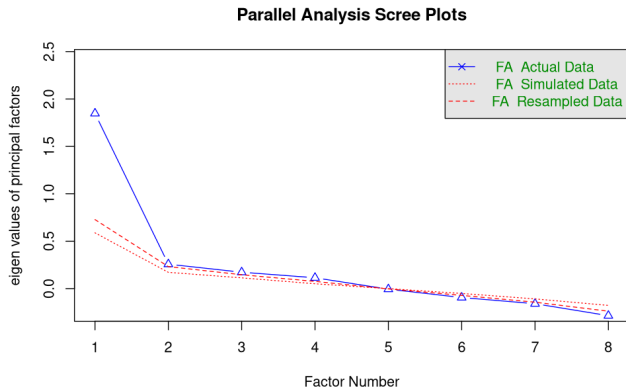
- ▶ Measurement: EFA → theory CFA; PCA as description
- ▶ Associations: ordinal, score, Bayesian, and nonlinear models
- ▶ Mechanisms: PLS paths, interaction, MGA
- ▶ Stress tests and prediction answer separate questions

What is reflective, formative, or observed?

- ▶ Tolerance: reflective hypothesis under examination
- ▶ Institutions and tangible costs: formative components
- ▶ CEO and ecological value: observed single items
- ▶ Item meanings constrain admissible models

EFA: the expected structure is not assured

- ▶ Ordinal polychorics, minres extraction, oblimin rotation
- ▶ Parallel analysis and identification must be read together



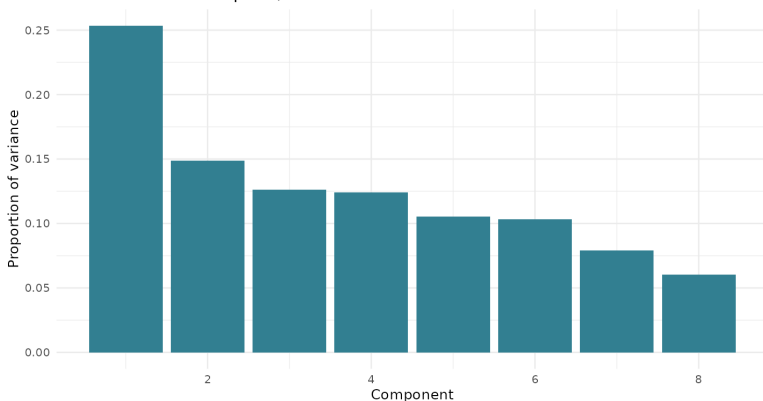
CFA: internal validation, not independent replication

- ▶ Three-item tolerance two-factor validation: CFI 0.967, RMSEA 0.048
- ▶ Village-disjoint development and validation partitions
- ▶ Weak word-item loadings and factor overlap remain

PCA describes variance; it does not validate constructs

- ▶ Standardized complete-case item PCA
- ▶ Loadings and explained variance supplied
- ▶ PCA components are not latent causes, IRT, or PLS-SEM

PCA: variance description, not construct validation



The HTMT question

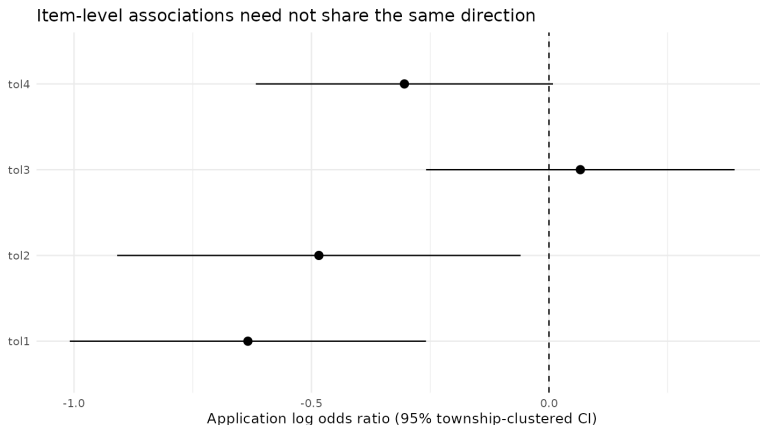
- ▶ Historical 1.112 must be reassessed after coding corrections
- ▶ Corrected reflective cost/tolerance diagnostics still show overlap
- ▶ HTMT is not a pass/fail rule for formative institutional contact

Primary estimands and uncertainty

- ▶ Four ordinal outcomes; Holm adjustment within each model family
- ▶ Unadjusted → background → resources → explanatory
- ▶ County effects; township-clustered uncertainty
- ▶ Report probabilities and intervals, not just p-values

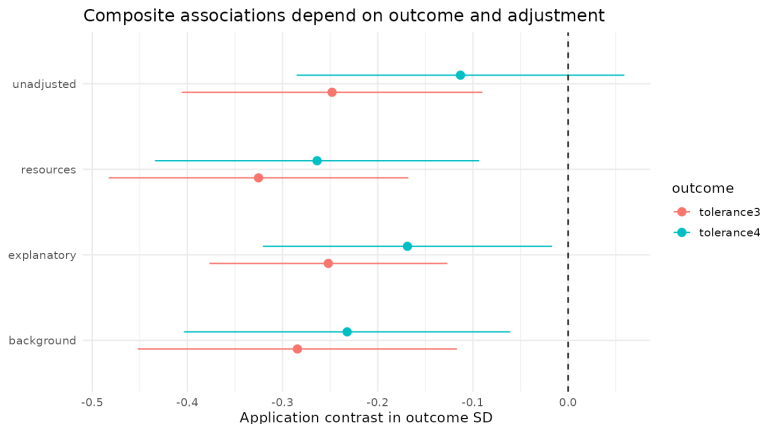
Item-level results differ

- ▶ The strongest background-adjusted signal concerns non-lethal management.



Composite results depend on the definition

- ▶ Three- and four-item scores are secondary summaries.



Ordinal assumptions and Bayesian checks

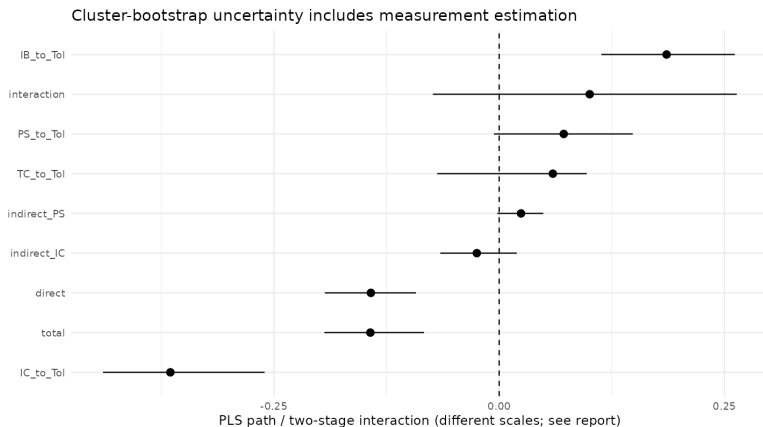
- ▶ 4/4 Bayesian models passed the prespecified MCMC checks
- ▶ Township random intercepts; weakly informative priors
- ▶ Nonparallel slopes and sparse-category fits remain concerns

PLS-SEM: compare specified models

- ▶ Baseline; mediation; conflict/prevention extension
- ▶ Three vs four tolerance items; IC1-only sensitivity
- ▶ Standard PLS-PM; full measurement refit in the bootstrap

Direct and indirect associations

- Intervals come from 5,000 county-stratified township resamples.



The proposed mechanisms remain uncertain

- ▶ Direct path: -0.142 [-0.193, -0.092]
- ▶ Indirect via costs: -0.025 [-0.065, 0.020]
- ▶ Indirect via institutions: 0.024 [-0.002, 0.049]
- ▶ These data do not demonstrate governance fatigue.

Small interaction; uncertain group differences

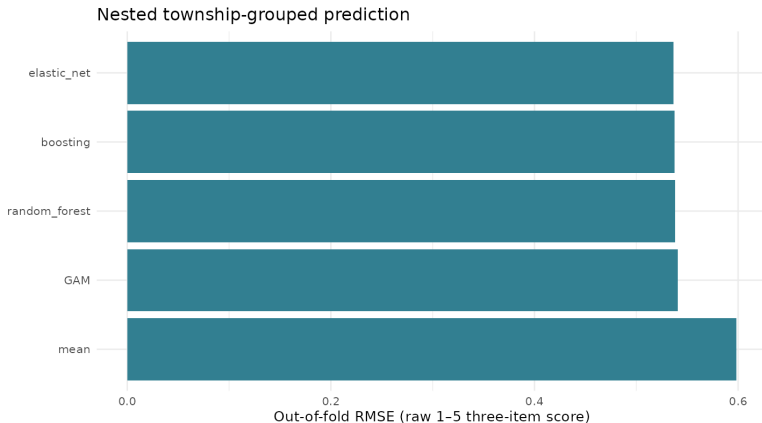
- ▶ Interaction: 0.100 [-0.074, 0.264]
- ▶ Interaction $f^2 = 0.003$
- ▶ All four cluster-bootstrap MGA intervals include zero
- ▶ No strong difference is not proof of equivalence

Secondary findings use more of the fieldwork

- ▶ Agricultural dependence and prevention deserve attention
- ▶ Applicant timeliness/satisfaction estimates are uncertain
- ▶ Insurance preferences, channels, crops, and seasons are profiled
- ▶ Secondary family uses BH multiplicity adjustment

How much can we predict?

- ▶ Best observed grouped-CV RMSE: 0.537 (elastic_net)

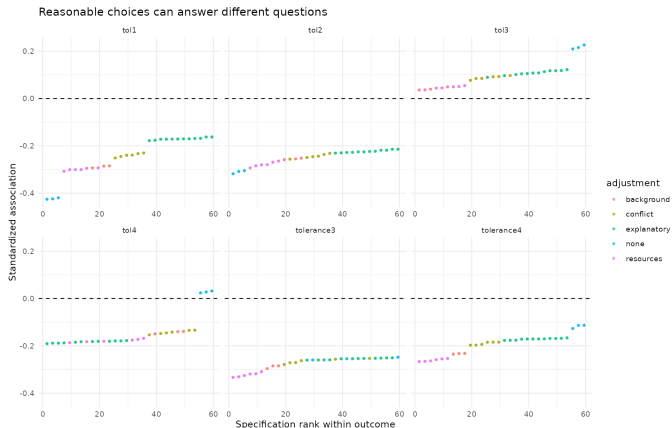


Cross-fitting checks adjusted associations

- ▶ Five AIPW contrasts: -0.216 to -0.193 raw score points
- ▶ Nuisance models predict unseen townships
- ▶ Overlap and balance are checked; causal assumptions remain

A transparent specification curve

- ▶ 360 distinct model/uncertainty combinations; outcomes shown separately



Stress-test the observations and assumptions

- ▶ Leave-one-township contrast range: -0.353 to -0.289
- ▶ Village clustering; county checks; extreme-loss sensitivity
- ▶ Unresolved-value handling and confounding-strength grid
- ▶ All failures and model choices remain visible

What can we defend?

- ▶ Applicants report lower tolerance on several management dimensions
- ▶ Measurement and outcome choice materially affect the story
- ▶ The proposed mediation and moderation are not established
- ▶ Policy failure is not identified by this survey

What remains for the final report?

- ▶ Clarify sampling, exclusions, payment timing, percentages, and units
- ▶ Resolve methodological questions from today's discussion
- ▶ Amend the decision register and rerun affected analyses
- ▶ Preserve the presentation release for comparison

Discussion

- ▶ Which tolerance dimensions matter for the policy question?
- ▶ What evidence would distinguish selection from policy effects?
- ▶ Which measurement interpretation is defensible?

Backup: Backup: direct answers to Mengxi

- ▶ Measurement: no clean, automatic latent structure
- ▶ HTMT: diagnose reflective blocks; do not merge formative behaviors blindly
- ▶ Moderation: small and uncertain; MGA: no strong difference
- ▶ Interpretation: association, not demonstrated causal mediation

Backup: Backup: selection and causal direction

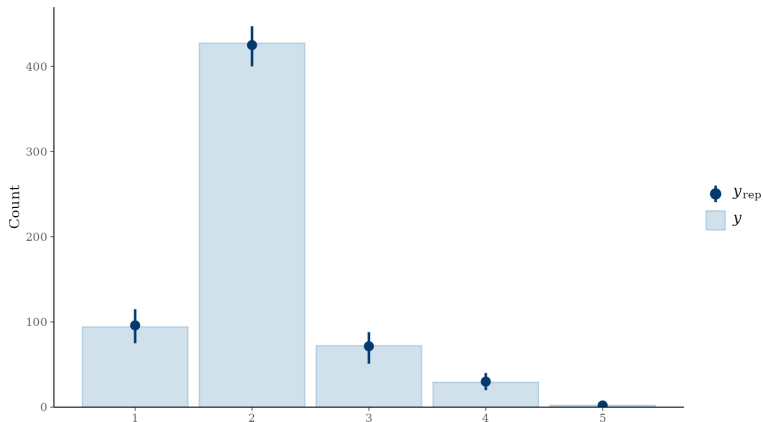
- ▶ Application is voluntary and measured cross-sectionally
- ▶ County, losses, attitudes, and institutions can shape selection
- ▶ Cross-fitting removes training reuse, not unmeasured confounding
- ▶ No valid instrument or longitudinal policy comparison is established

Backup: Backup: why not interpret nonsignificance as equality?

- ▶ Group differences can be imprecise
- ▶ Equivalence needs a justified margin and adequate precision
- ▶ Group-standardized coefficients depend on variance
- ▶ MICOM non-rejection is not proof of full invariance

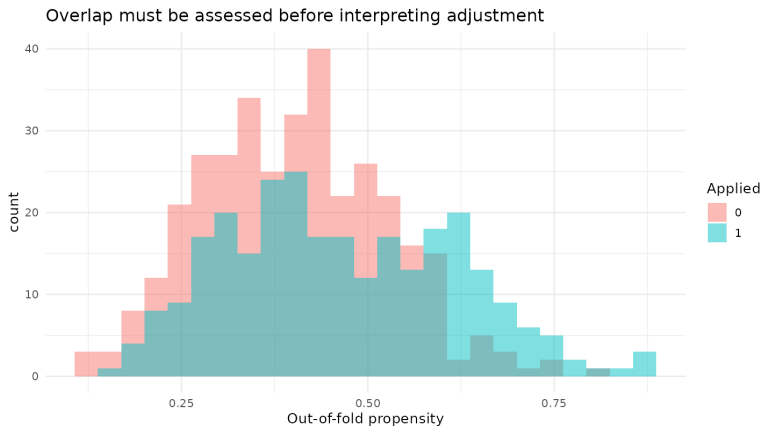
Backup: Backup: posterior predictive check

- MCMC diagnostics and model fit answer different questions.



Backup: Backup: propensity overlap

- Inspect support, balance, clipping, and effective sample size.



Backup: Backup: p-hacking safeguards

- ▶ Dated plan; prior inspection disclosed
- ▶ Full results and failures; no significance-driven deletion
- ▶ Deduplicated specifications; multiplicity; internal validation
- ▶ No p-curve or certificate of 'no p-hacking'

Backup: Backup: missingness and anomalies

- ▶ Structural payment missingness is never imputed
- ▶ Word zeros are absent answers, not positive sentiment
- ▶ Percentages above 100% are flagged, not guessed
- ▶ Median/mode sensitivity is limited, not full multiple imputation

Backup: Backup: reproducibility

- ▶ R targets pipeline and renv lockfile
- ▶ Source/data hashes, seeds, grouped folds, model objects
- ▶ Reports and slides read the same saved tables
- ▶ One build command; separate render and validation commands

Backup: Backup: methodological references

- ▶ lavaan: categorical WLSMV; psych: EFA and parallel analysis
- ▶ cSEM: PLS-PM and MICOM; brms: ordinal multilevel models
- ▶ Simonsohn et al. (2020): specification curve analysis
- ▶ DoubleML: cross-fitting and sensitivity assumptions